AWS VPC Peering

Designing and validating private connectivity between two Amazon Virtual Private Clouds, with secure bastion access, bidirectional testing, controlled failure and recovery, and complete resource cleanup.

Subhabrata Pal

Submission date: 30 October 2026

AT A GLANCE

01 / Report guide and key results

Use this guide to find the network design, configuration, test results and cleanup records. Evidence labels E01-E30 match the captioned screenshots on pages 9-38.

4 / 4
Forward ping replies

4 / 4
Reverse ping replies

4 / 4
Recovery replies

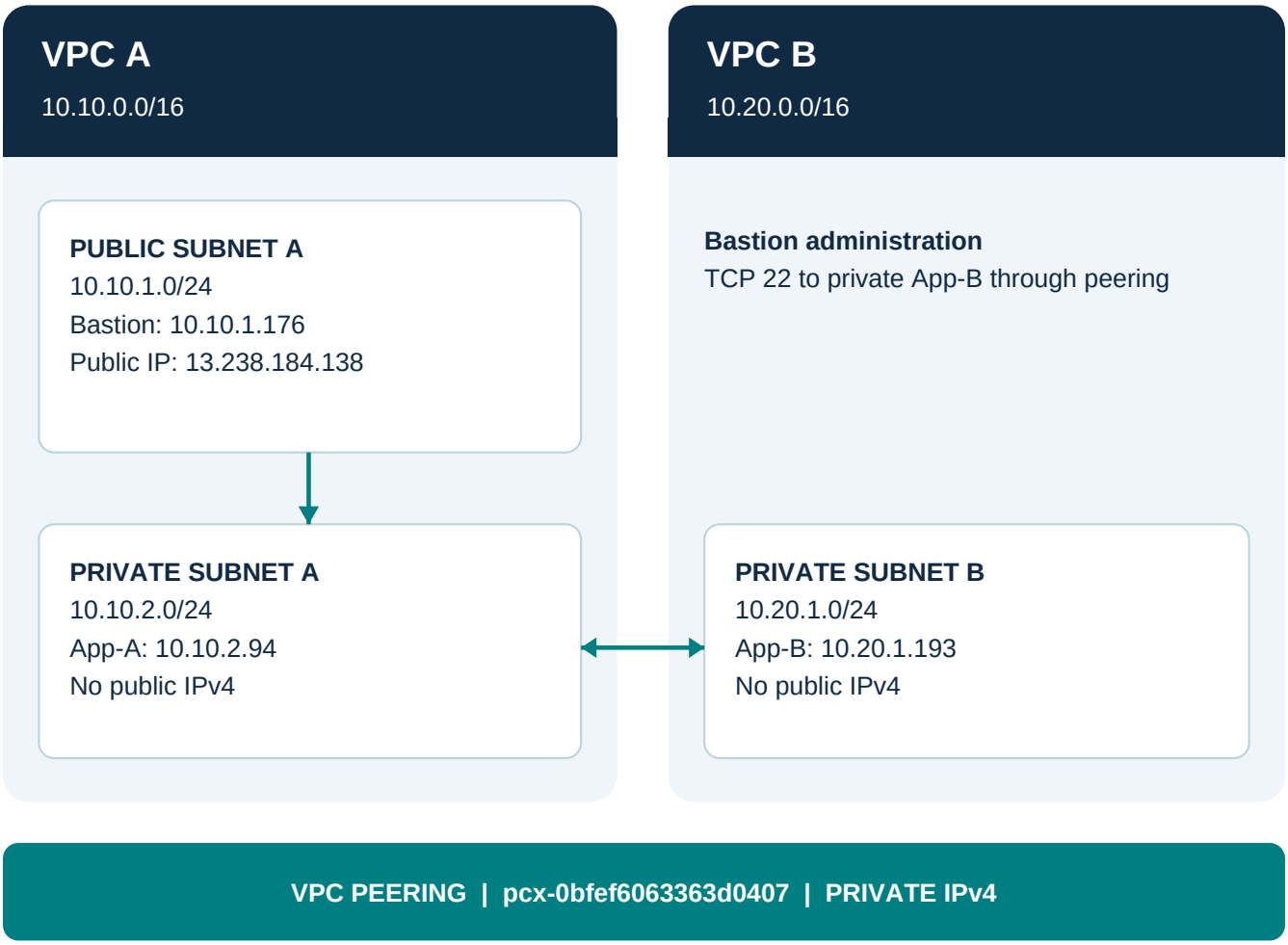
Section / what you will find	Report page	Screenshots
Architecture - VPCs, subnets and traffic paths	3	E01-E02, E06-E07
Inventory - actual resource IDs and private IPs	4	E01-E05, E14-E16
Routing - all three tables and associations	5	E02, E07-E10
Security - inbound rules and bastion access	6	E11-E15, E18
Validation - two-way ping, SSH and HTTP	7	E14-E19
Controlled failure - loss and recovery	7	E20-E21
Cleanup - removed resources and learning	8	E22-E30
Evidence atlas - all 30 original screenshots	9-38	E01-E30

How to read the evidence: each screenshot has an E-number, a descriptive heading, a concise result caption. Evidence E01 is on page 9; the remaining captures follow in sequence.

Verification notes: no before-peering baseline screenshot was supplied. Final ICMP rule details, snapshot/NAT inventories, local key removal and Billing checks are not independently captured. Observed results and user-reported details are distinguished throughout.

02 / Implemented architecture

Sydney | ap-southeast-2 | Two private application hosts



Private application path. App-A sends traffic to App-B through the 10.20.0.0/16 peering route. App-B returns it through the 10.10.0.0/16 route. Both hosts use private addresses and have no public IPv4.

Administration path. The bastion is the public entry point in VPC A. It opens SSH connections to App-A over the local VPC route and to App-B through the peer. Security groups determine which connections are accepted.

Internet access and isolation. VPC A has an attached internet gateway for its public subnet. E06 also shows an extra gateway on VPC B; App-B remains on a private table with no internet-default route (E10). Peering does not share an internet gateway between VPCs.

03 / Actual resource inventory

Resource	Actual identifier / addressing
VPC A - ewb-vpc-a	vpc-09d5f914c63cc5a49 10.10.0.0/16
VPC B - ewb-vpc-b	vpc-0786740c8561cb75a 10.20.0.0/16
Public A - ewb-a-public	subnet-0678828ab2ca58707 10.10.1.0/24
Private A - ewb-a-private	subnet-05b29203eb817f6a1 10.10.2.0/24
Private B - ewb-b-private	subnet-0229ef2a8f19098b7 10.20.1.0/24
Bastion - ewb-bastion	i-0d76e7d61e29e264d 10.10.1.176 / public 13.238.184.138
App-A - ewb-private-a	i-0e0e01bb22692993b 10.10.2.94 / no public IPv4
App-B - ewb-private-b	i-0a2fd94df4ebe0d2d 10.20.1.193 / no public IPv4
Compute / key pair	t3.micro; ap-southeast-2a; Amazon Linux 2023 Key pair name: ewb-key

Sources: E01-E05 and E14-E16. VPC ranges do not overlap and each /24 belongs to its parent /16. All three instances were running with 3/3 status checks passed in E03. Screenshots expose a key filename only; no private-key contents are included.

04 / Peering and routing

Active link: pcx-0bfef6063363d0407, Name tag ewb-vpc-a-to-b. Requester: VPC A; accepter: VPC B. Both endpoints are in Sydney and the same account (E07).

Route table / explicit association	Observed routes
ewb-a-public-rt rtb-0c1b1c5e37837abfb Public A subnet	10.10.0.0/16 > local 0.0.0.0/0 > igw-0c2c4e748f50f5886 10.20.0.0/16 > peering
ewb-a-private-rt rtb-042d57fd40f186487 Private A subnet	10.10.0.0/16 > local 10.20.0.0/16 > peering
ewb-rt-b-private rtb-0152682c054073e53 Private B subnet	10.20.0.0/16 > local 10.10.0.0/16 > peering

Forward path. App-A sends to 10.20.1.193. Private A's table selects 10.20.0.0/16 through the active peer; the destination security group must allow the request.

Return path. App-B replies to 10.10.2.94 through its 10.10.0.0/16 peering route. Return routing is essential even when security groups allow the traffic.

Bastion path. The public A table also needs 10.20.0.0/16 through the peer for SSH to App-B. That specific route wins over 0.0.0.0/0. App-B receives the bastion's private source address, 10.10.1.176.

E08-E10 show the route rows and explicit subnet associations, matched to E02. E06 shows igw-0c2c4e748f50f5886 attached to A and extra igw-0f55f4f601d7535a4 attached to B. Peering provides neither transitive routing nor shared internet-gateway access.

05 / Security and administration

Security group	Inbound rules visible in screenshot
ewb-sg-bastion sg-096d86d8fa5e42712	SSH / TCP 22 from 223.184.142.250/32 (E11)
ewb-sg-private (App-A) sg-084ed6814b42e3caf	SSH / TCP 22 from the bastion security group (E12)
ewb-sg-app-b sg-0e572bf7b9f5b4552	SSH / TCP 22 from 10.10.0.0/16 HTTP / TCP 80 from 10.20.0.0/16 HTTP / TCP 80 from 10.10.1.176/32 (E13)

Observed access scope. App-A allows SSH through a bastion security-group reference. App-B permits SSH from the full VPC A CIDR, which is broader than a single-host /32 rule. Host-specific sources would provide tighter access control.

ICMP evidence gap. E12-E13 do not display the later ICMP rules needed for the successful ping tests. E17-E21 prove the test outcomes, but the final ICMP source, type/code and the console rule removal/restoration were not captured. Each SG shows one outbound permission entry; its details are not expanded.

Private SSH. E14 shows a local OpenSSH ProxyCommand through 13.238.184.138 to App-A; hostname -I returns 10.10.2.94. E18 shows the same bastion method to App-B and a remote shell at 10.20.1.193. E15 independently identifies App-B.

The command references a local .pem path. It contains no key material. The bastion opens the destination connection; the private hosts remain without public IPv4 addresses (E04-E05).

06 / Validation and controlled failure

Test / endpoint	Observed result / evidence
Before-peering baseline	Not supplied. The later failure is not a baseline.
Forward: App-A > 10.20.1.193	4 sent / 4 received; 0% loss; avg 0.270 ms (E17)
Reverse: App-B > 10.10.2.94	4 sent / 4 received; 0% loss; avg 0.191 ms (E18)
Private SSH via bastion	App-A shell (E14); App-B shell (E18)
Private application address check	10.10.2.94 (E14); 10.20.1.193 (E15)
Controlled failure: App-A > App-B	4 sent / 0 received; 100% loss (E20)
Recovery: App-A > App-B	4 sent / 4 received; 0% loss; avg 0.241 ms (E21)
Additional HTTP probe	curl -I http://10.20.1.193 returns HTTP/1.0 200 OK (E19)

All ping tests use ping -c 4 and the peer's private IP. The failure screenshot is labelled ICMP blocked, and the supplied project details attribute it to a controlled ICMP change. The terminal proves loss and restoration; it does not independently prove which security rule was changed.

The HTTP response identifies SimpleHTTP/0.6 with Python/3.9.25. The screenshot does not identify the client host, so it is evidence of an HTTP response only, not a proven App-A-to-App-B HTTP test.

Active peering alone is insufficient evidence of connectivity: the route tables, return path and traffic permissions must work together. The positive tests and deliberate outage illustrate that dependency without inventing a missing pre-peering capture.

07 / Cleanup and lessons learned

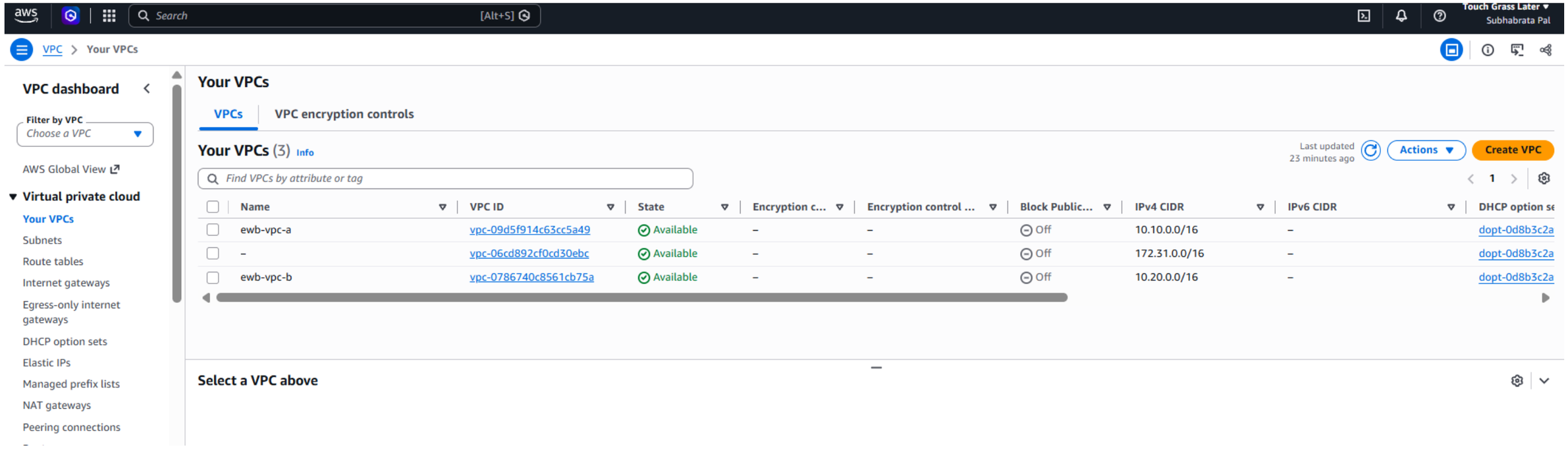
Cleanup item	Evidence and status
Three EC2 instances	All show Terminated (E22).
EBS / Elastic IP	No volumes (E23); no Elastic IP allocations (E24).
Peering and peer routes	Deletion banner; 4 of 4 related routes deleted (E25).
Security groups	Four SGs deleted; default groups remain in that capture (E26).
Project subnets	Three project subnet IDs deleted; default subnets remain (E27).
Internet gateways	Only default VPC gateway remains; deletion banner (E28).
Project VPCs	Only default 172.31.0.0/16 VPC remains (E29).
EC2 key pair	One key pair deleted; no key pairs displayed (E30).
Snapshots / NAT gateways	Empty per user report; no dedicated screenshot in archive.
Local key / Billing check	Not independently verified in supplied evidence.

Lessons learned. Plan non-overlapping CIDRs first. Map each subnet to its actual route table and check both directions. The bastion needs its own peer route for private SSH to App-B. Keep inbound rules narrow, preserve a record of the final ICMP rules, and capture baseline evidence before changing connectivity.

Save evidence before teardown and check dependent resources after termination. Cleanup screenshots establish the displayed regional state; they do not independently establish the final account bill. The default VPC was retained.

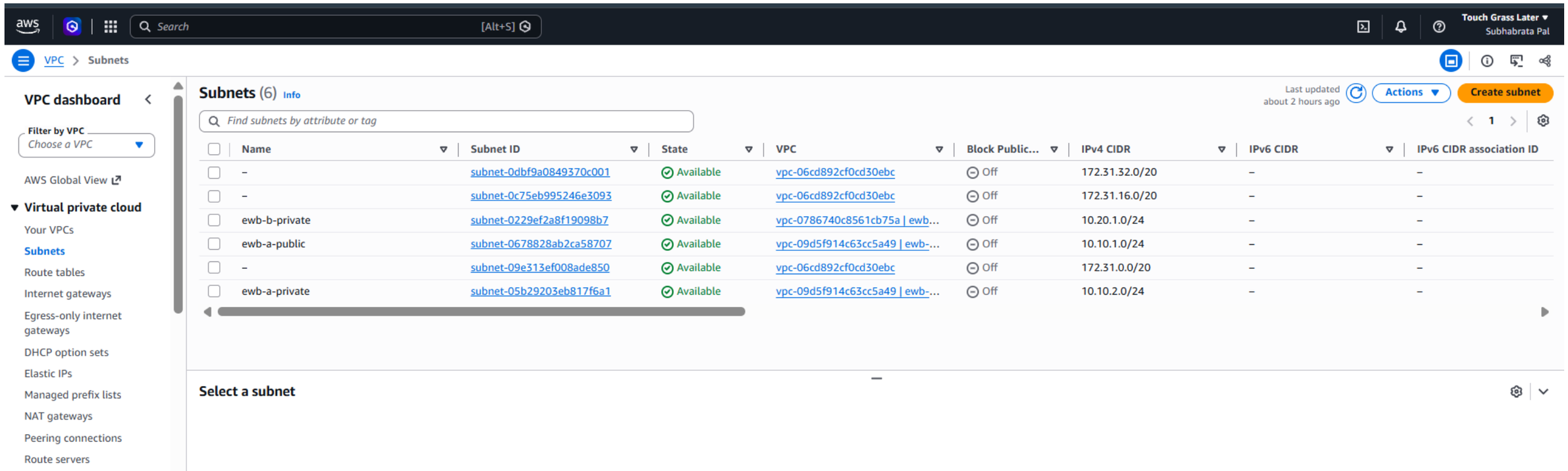
E01 / VPC inventory

VPC A and VPC B are available with non-overlapping CIDRs; the default VPC is also listed.



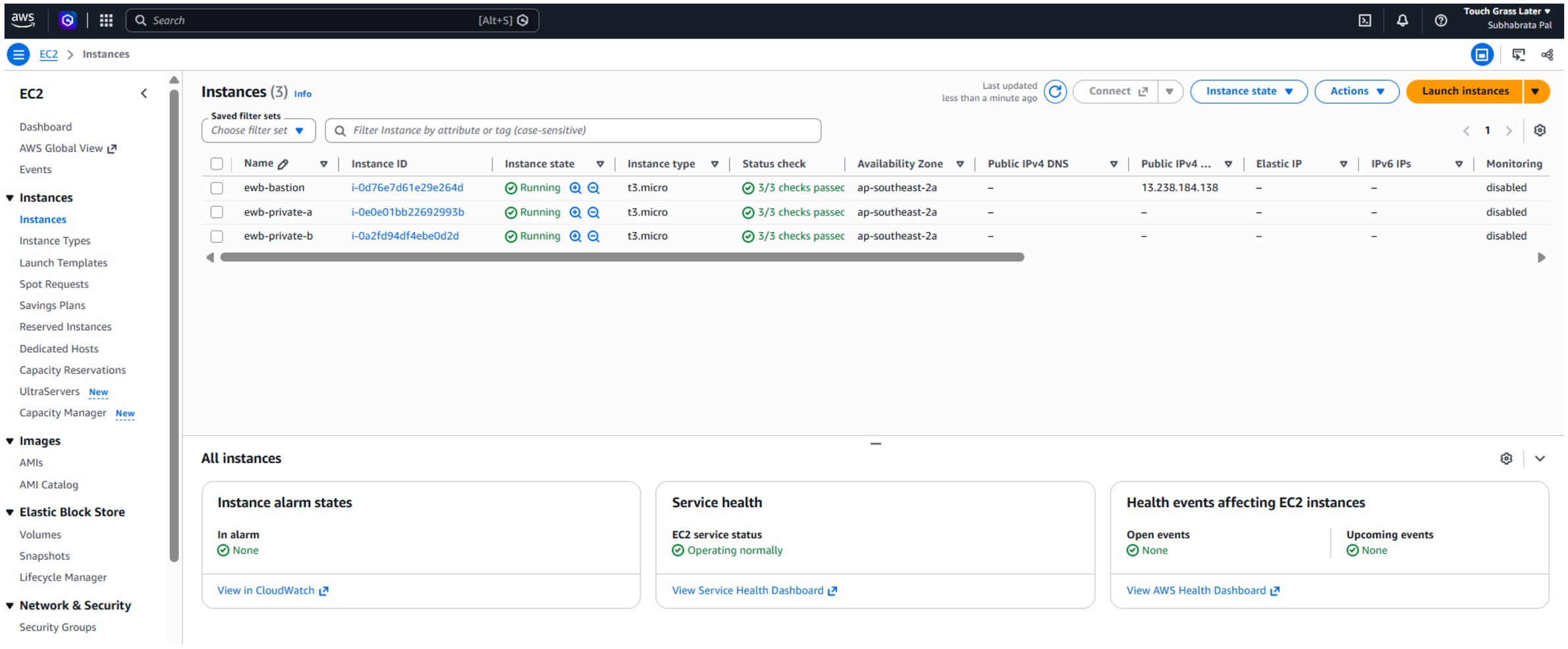
E02 / Subnet inventory

Three project subnets and their actual CIDRs, subnet IDs and parent VPCs.



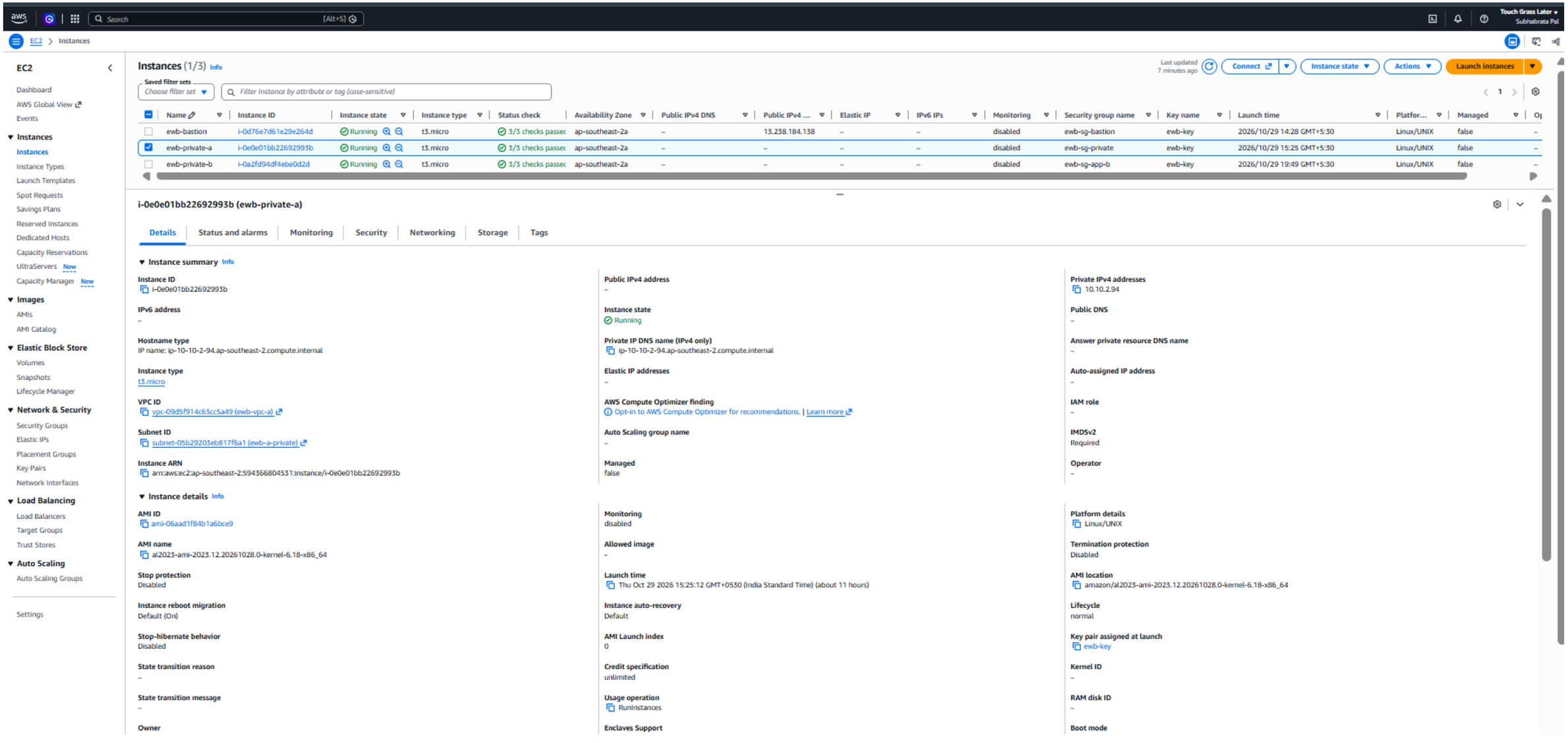
E03 / EC2 inventory - running

Bastion, App-A and App-B are running; t3.micro, ap-southeast-2a and 3/3 checks passed.



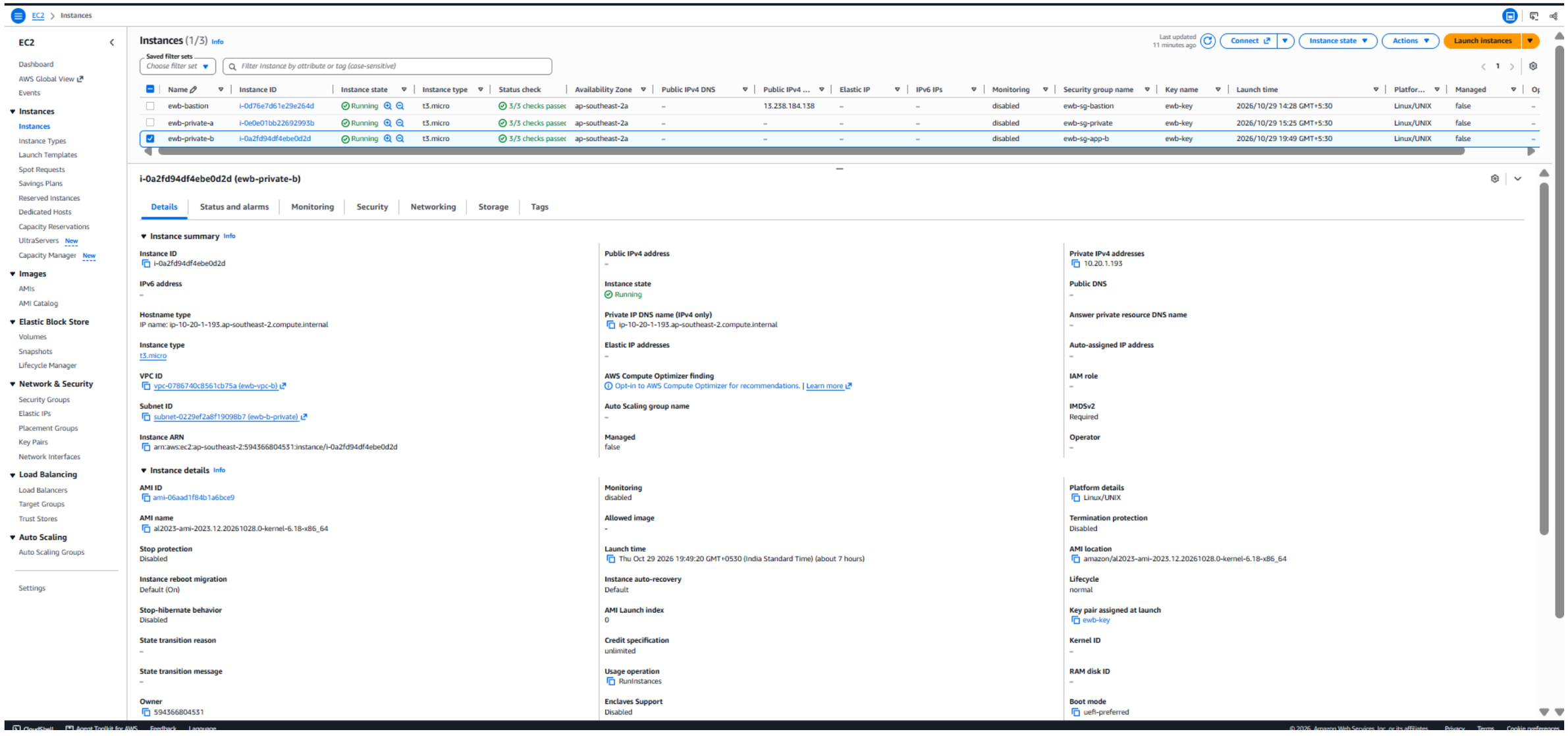
E04 / App-A - private addressing

App-A details show 10.10.2.94, its VPC/subnet IDs and no public IPv4.



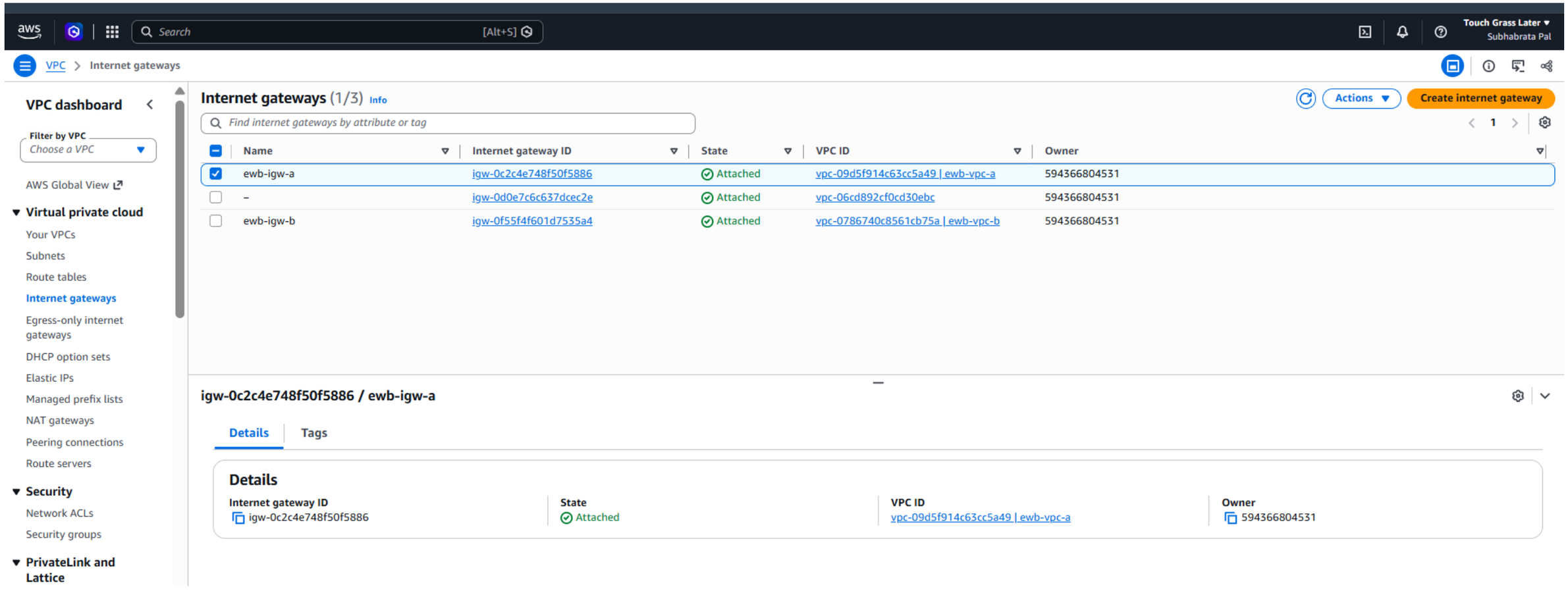
E05 / App-B - private addressing

App-B details show 10.20.1.193, its VPC/subnet IDs and no public IPv4.



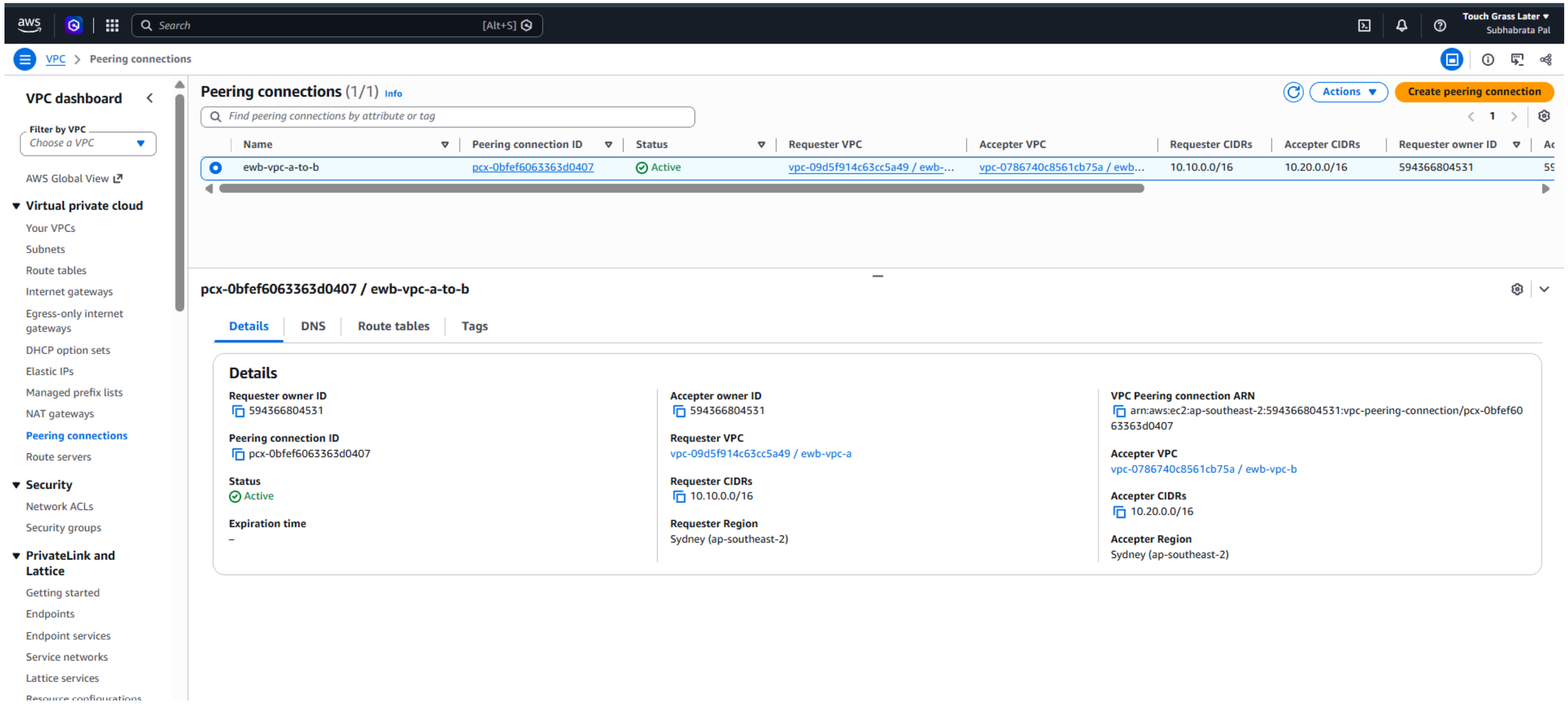
E06 / Internet gateway attachment

A gateway is attached to VPC A. An extra VPC B gateway is visible and recorded as a deviation.



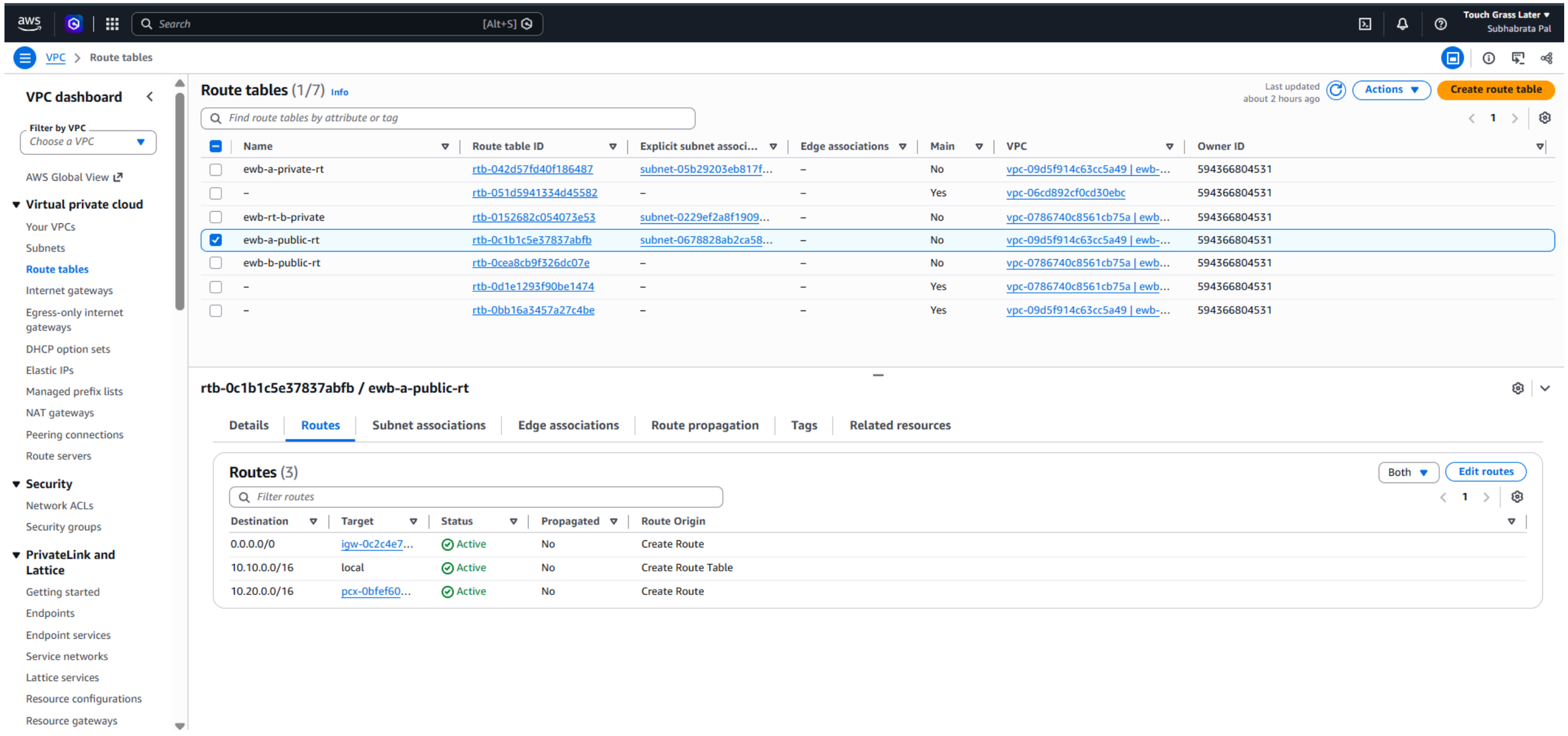
E07 / Peering connection - Active

The active peer connects requester VPC A and accepter VPC B in Sydney.



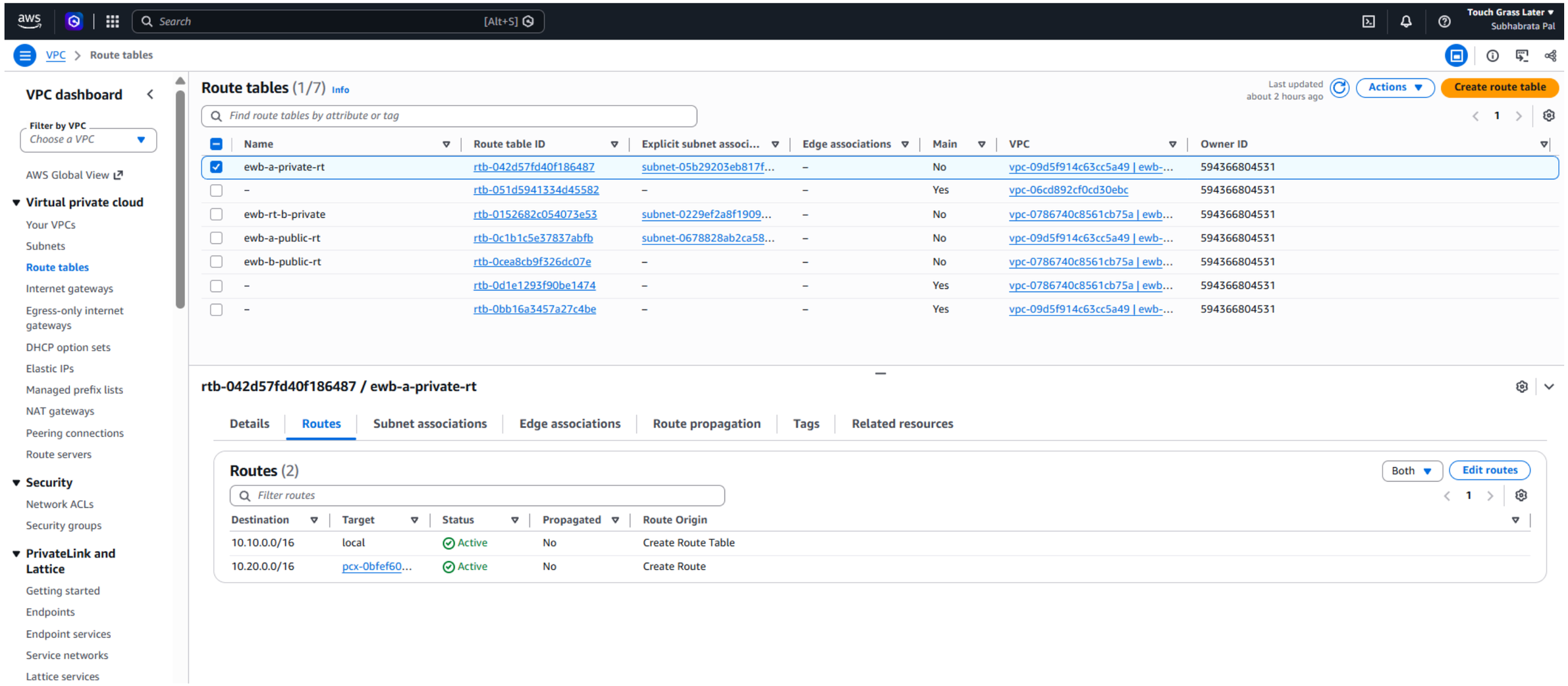
E08 / Public A - routes and association

Local, internet-default and peer routes support bastion administration.



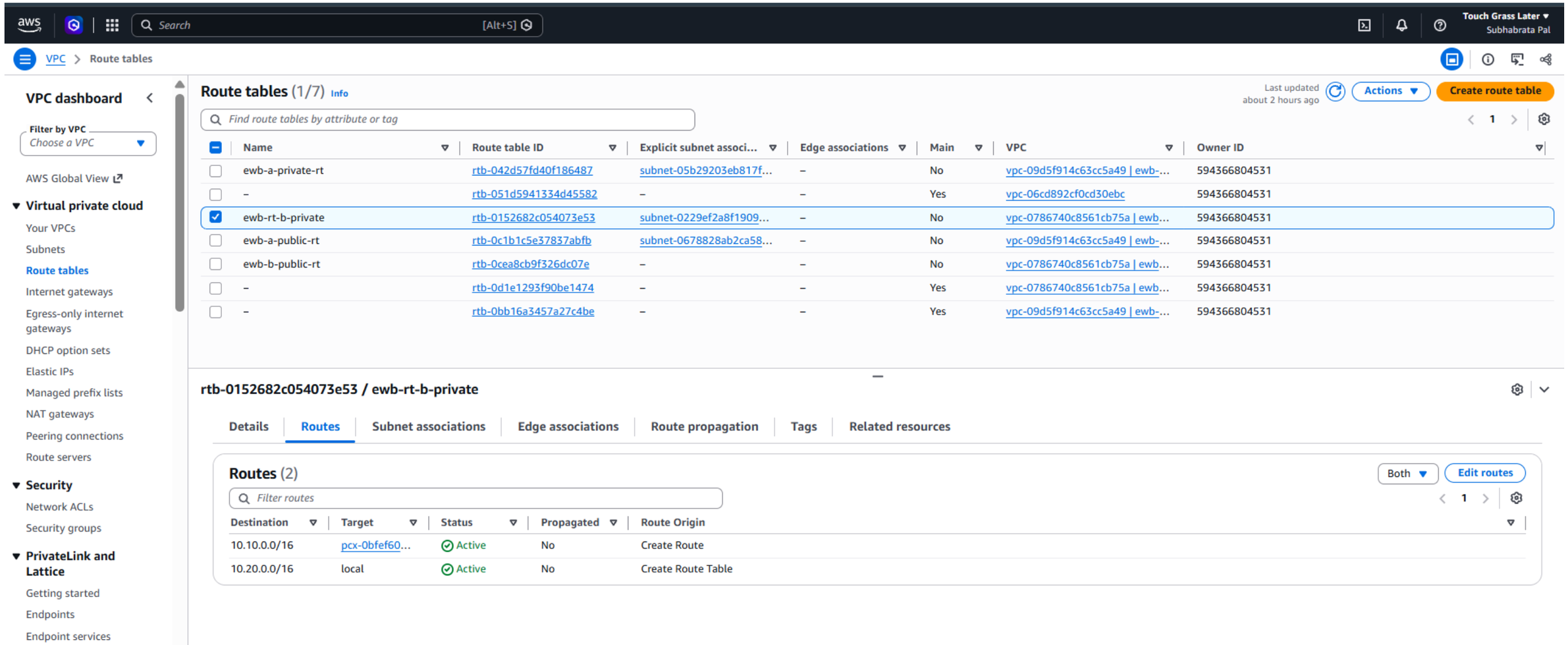
E09 / Private A - routes and association

Local A route plus 10.20.0.0/16 through the peer; no internet-default route shown.



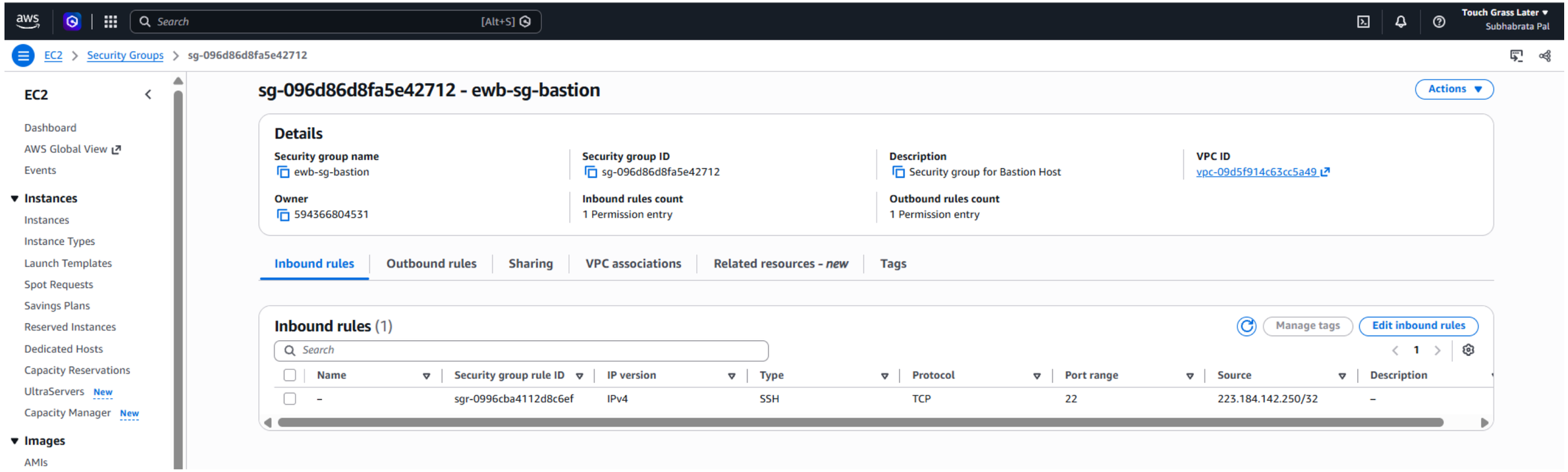
E10 / Private B - routes and association

Local B route plus 10.10.0.0/16 through the peer provide the return path.



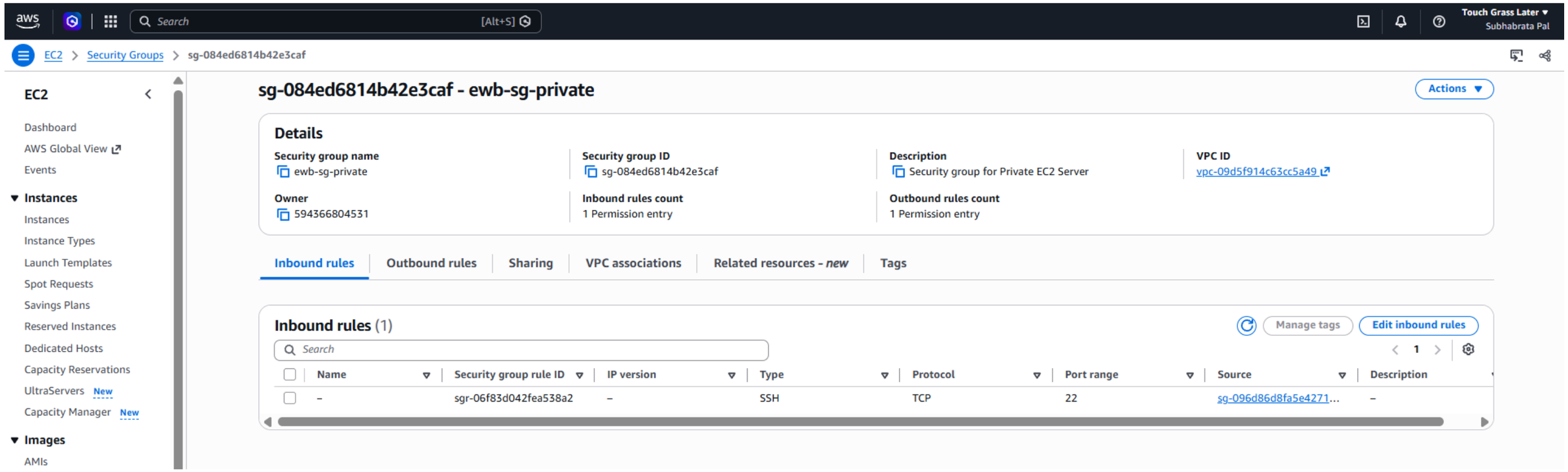
E11 / Bastion - inbound SSH rule

TCP 22 is scoped to the displayed operator public IPv4 /32.



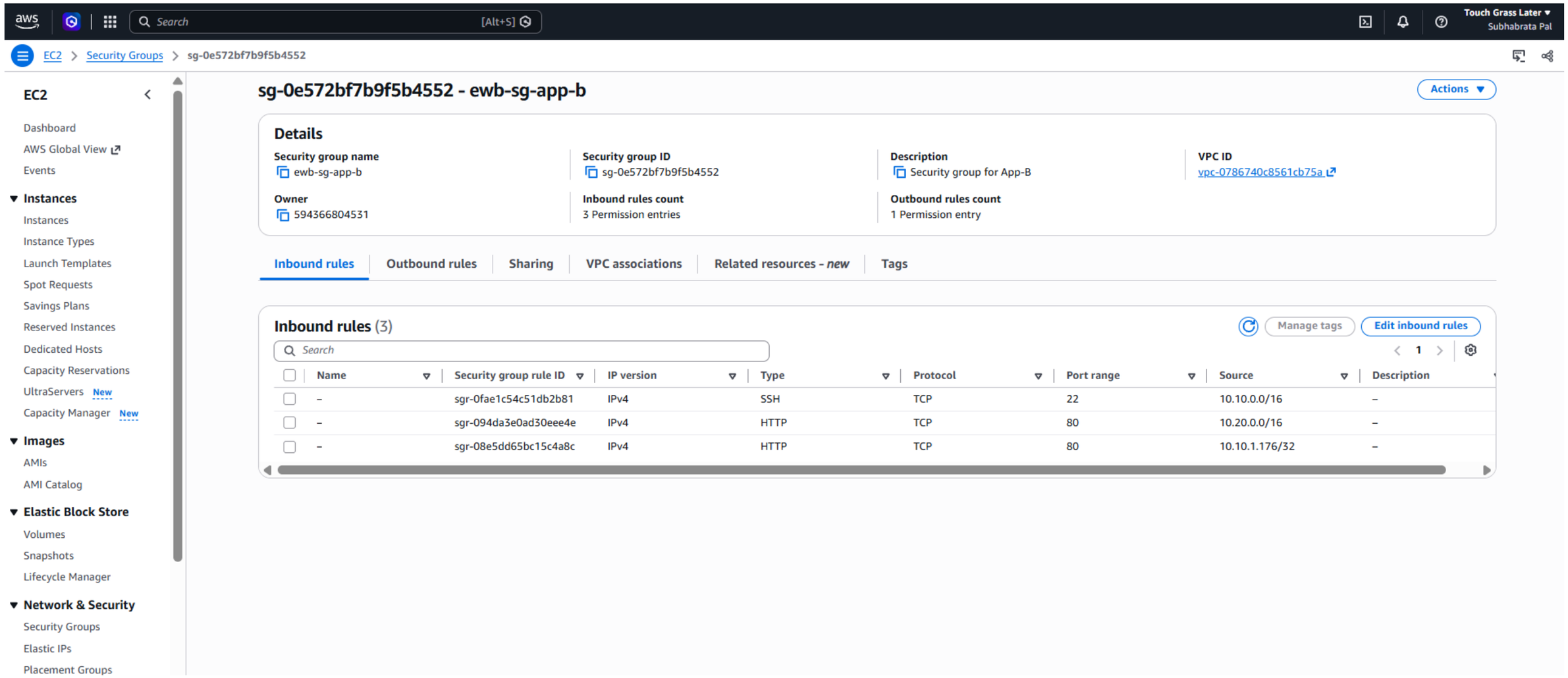
E12 / App-A - captured inbound rule

SSH is allowed from the bastion SG. This capture does not show final ICMP permissions.



E13 / App-B - captured inbound rules

SSH from VPC A CIDR and two HTTP rules are visible. Final ICMP permissions are not shown.



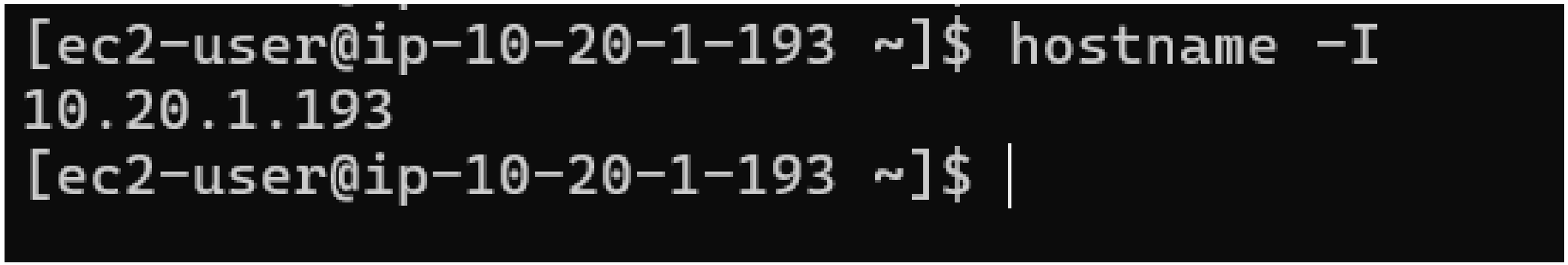
E14 / App-A - SSH through bastion succeeds

ProxyCommand reaches App-A; hostname -I confirms 10.10.2.94.

```
PS C:\Users\subha> ssh -i "C:/Users/subha/OneDrive/Desktop/ewb-key.pem" -o "
ProxyCommand=ssh -i C:/Users/subha/OneDrive/Desktop/ewb-key.pem -W %h:%p ec2
-user@13.238.184.138" ec2-user@10.10.2.94
       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
Last login: Thu Oct 29 13:16:36 2026 from 10.10.1.176
[ec2-user@ip-10-10-2-94 ~]$ hostname -I
10.10.2.94
[ec2-user@ip-10-10-2-94 ~]$
```

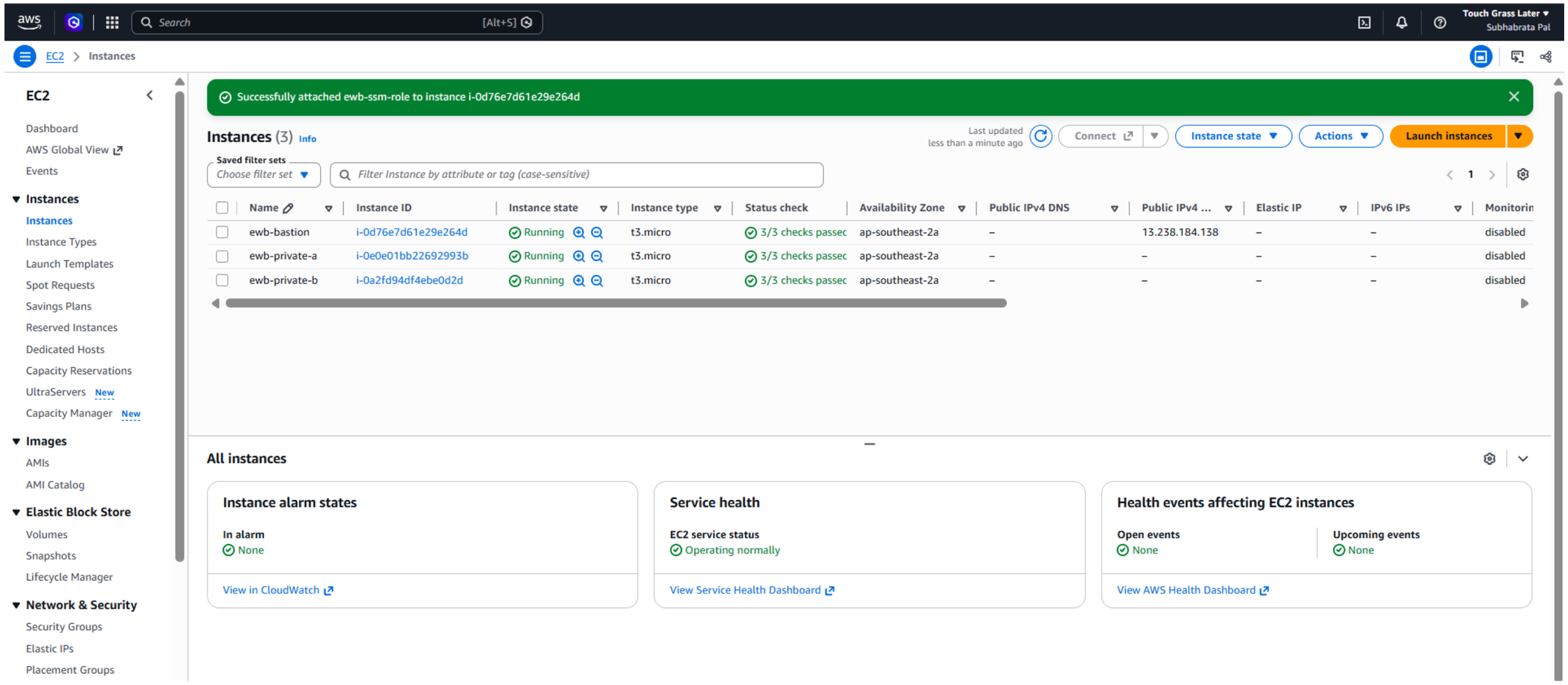
E15 / App-B - private host identification

The App-B shell returns 10.20.1.193 from hostname -I; the full bastion command appears in E18.



E16 / EC2 inventory - additional original capture

All three instances are running. A banner reports an IAM role attachment; its permissions are not shown.



E17 / App-A > App-B - forward ping succeeds

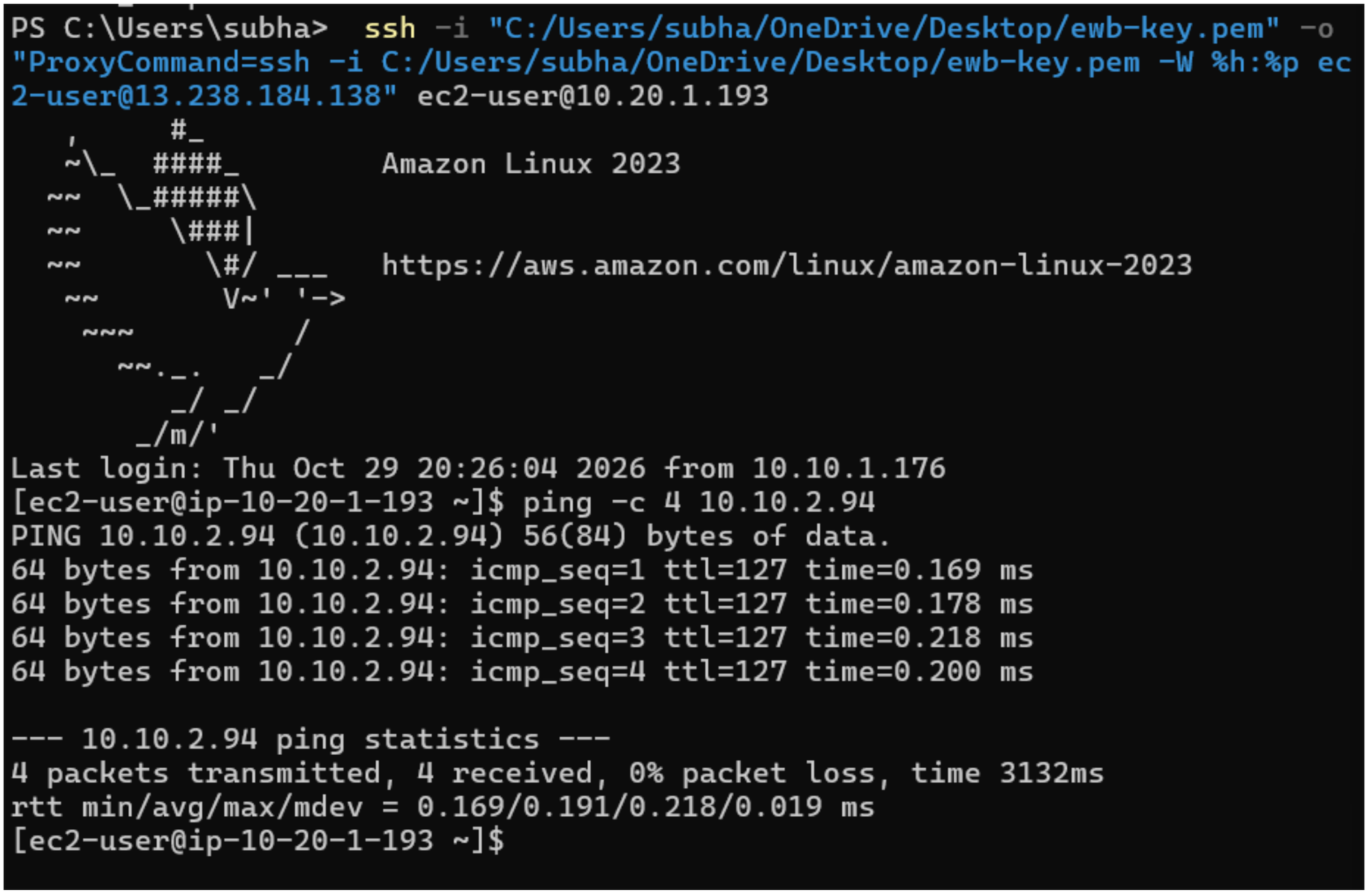
10.10.2.94 sends to 10.20.1.193: 4/4 replies, 0% loss.

```
[ec2-user@ip-10-10-2-94 ~]$ ping -c 4 10.20.1.193
PING 10.20.1.193 (10.20.1.193) 56(84) bytes of data.
64 bytes from 10.20.1.193: icmp_seq=1 ttl=127 time=0.214 ms
64 bytes from 10.20.1.193: icmp_seq=2 ttl=127 time=0.300 ms
64 bytes from 10.20.1.193: icmp_seq=3 ttl=127 time=0.264 ms
64 bytes from 10.20.1.193: icmp_seq=4 ttl=127 time=0.304 ms

--- 10.20.1.193 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3123ms
rtt min/avg/max/mdev = 0.214/0.270/0.304/0.036 ms
[ec2-user@ip-10-10-2-94 ~]$
```

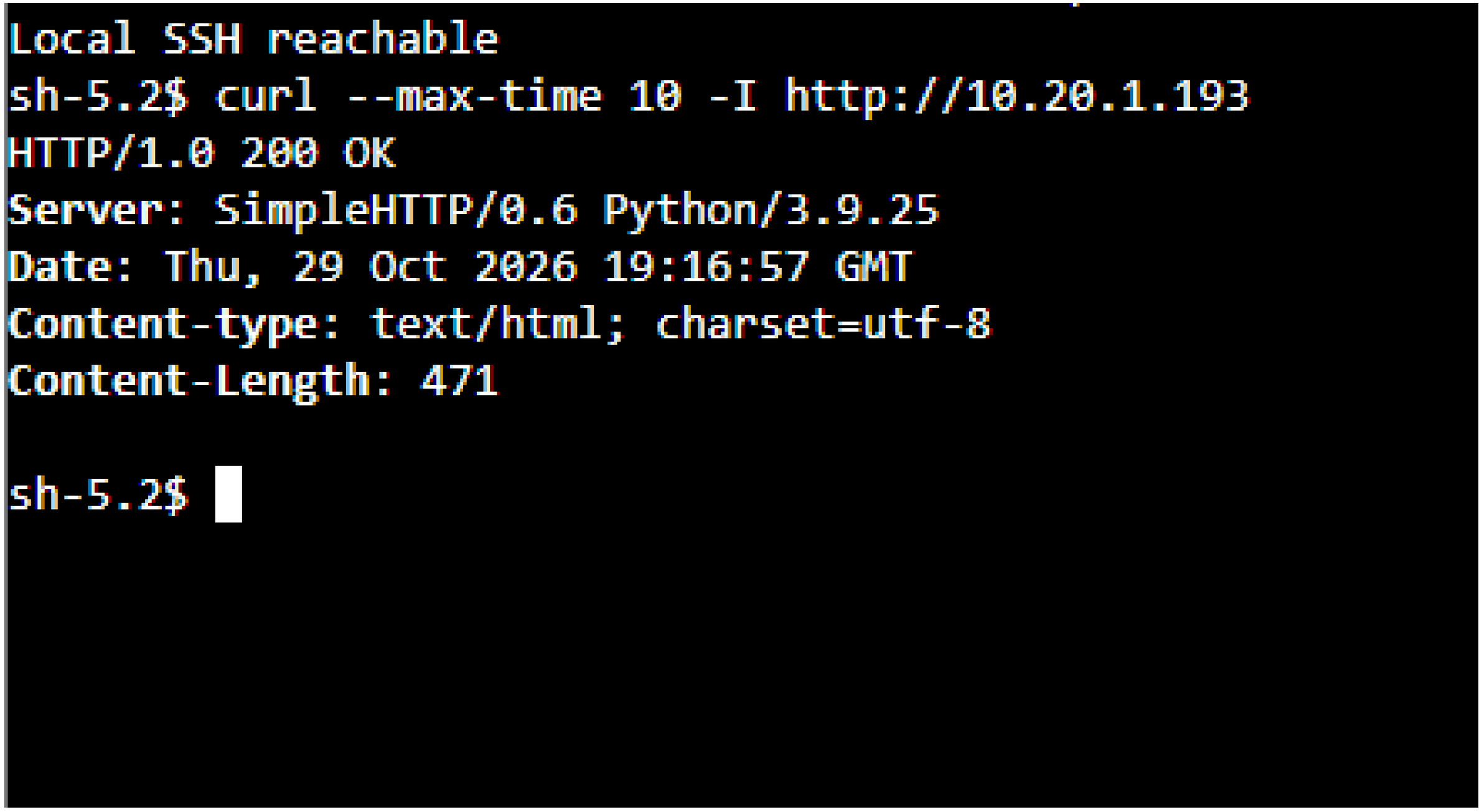
E18 / App-B > App-A - reverse ping succeeds

Local SSH through the bastion reaches App-B; reverse ping to 10.10.2.94 receives 4/4 replies.



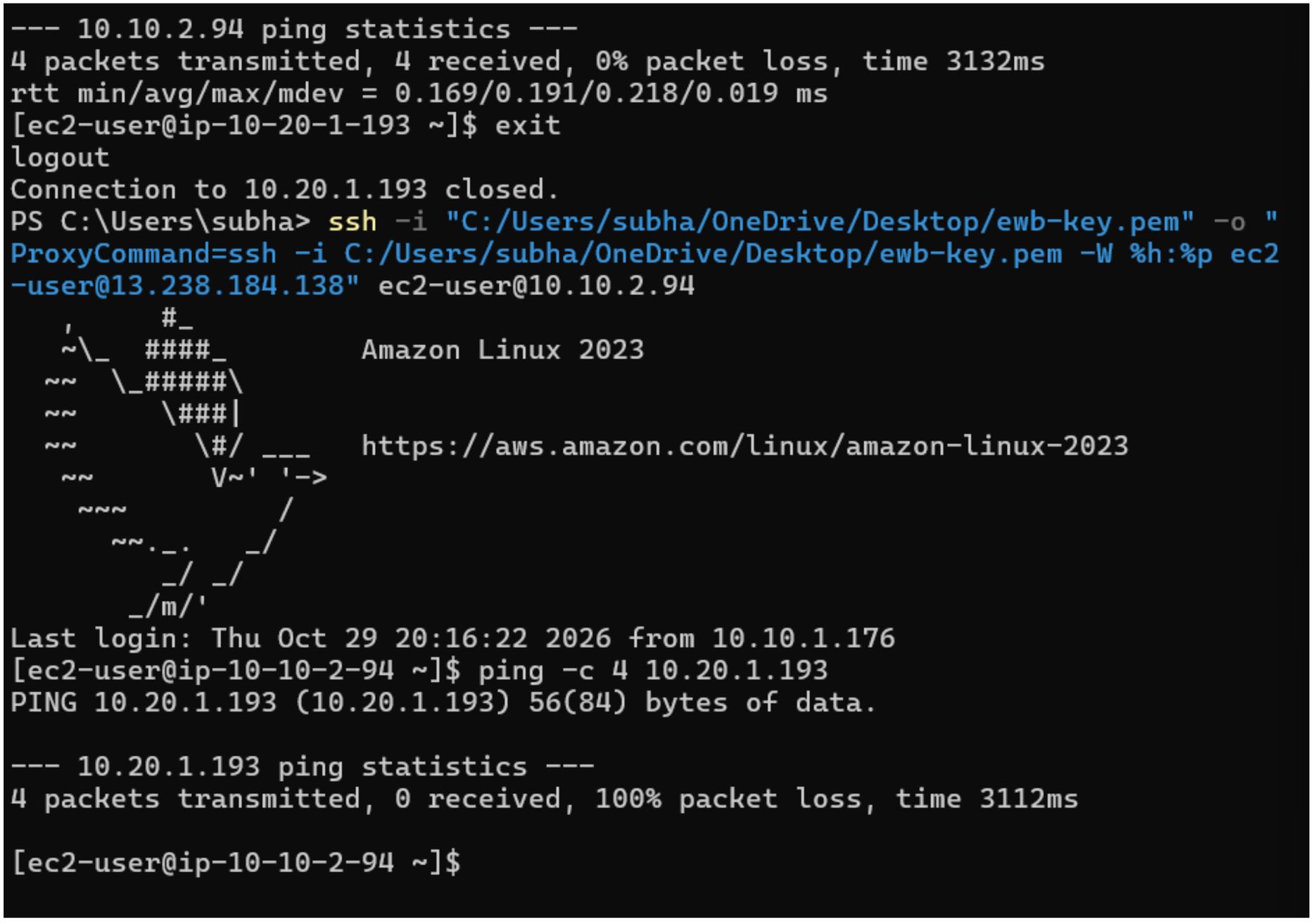
E19 / App-B endpoint - HTTP response succeeds

The private endpoint returns HTTP/1.0 200 OK. The client host is not established by this capture.



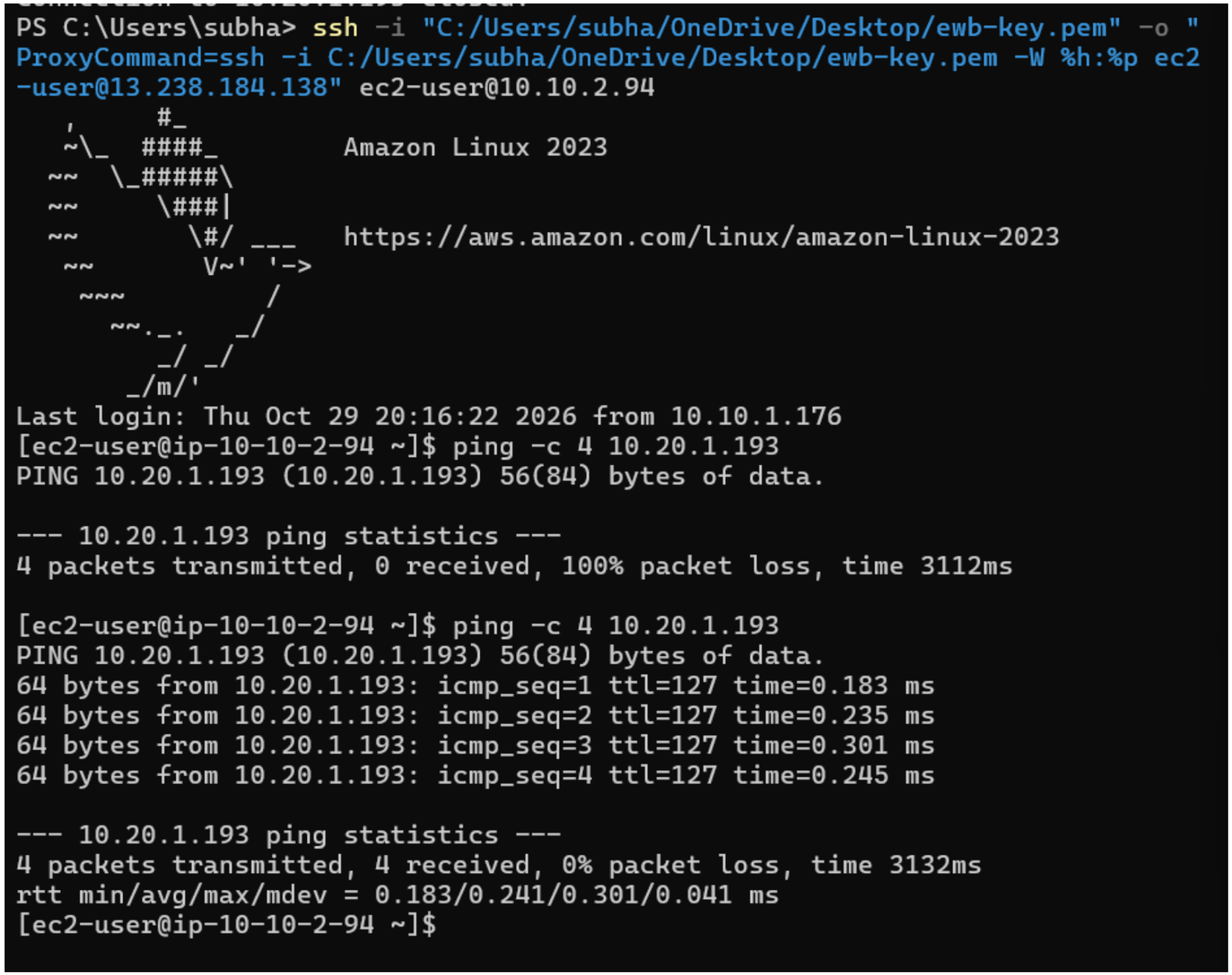
E20 / App-A > App-B - controlled ping failure

The later ping to 10.20.1.193 receives 0/4 replies. This is not evidence of the pre-peering baseline.



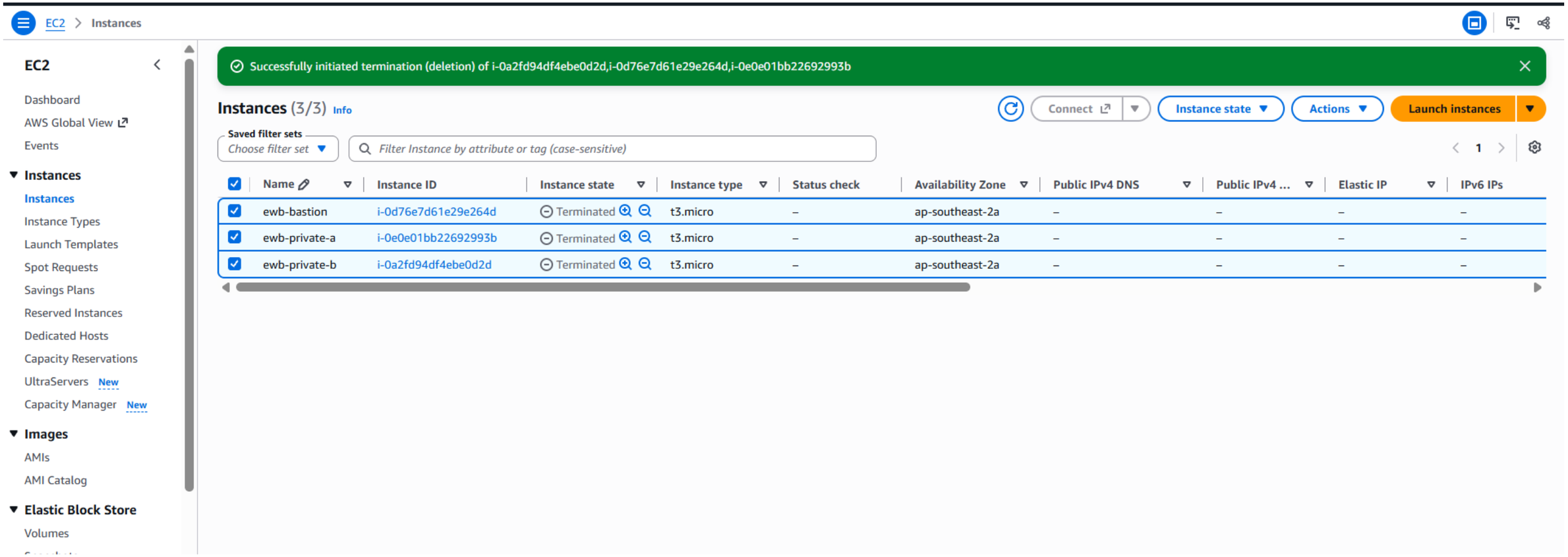
E21 / App-A > App-B - recovery succeeds

A failed run is followed by 4/4 replies after recovery. The console rule change is not pictured.



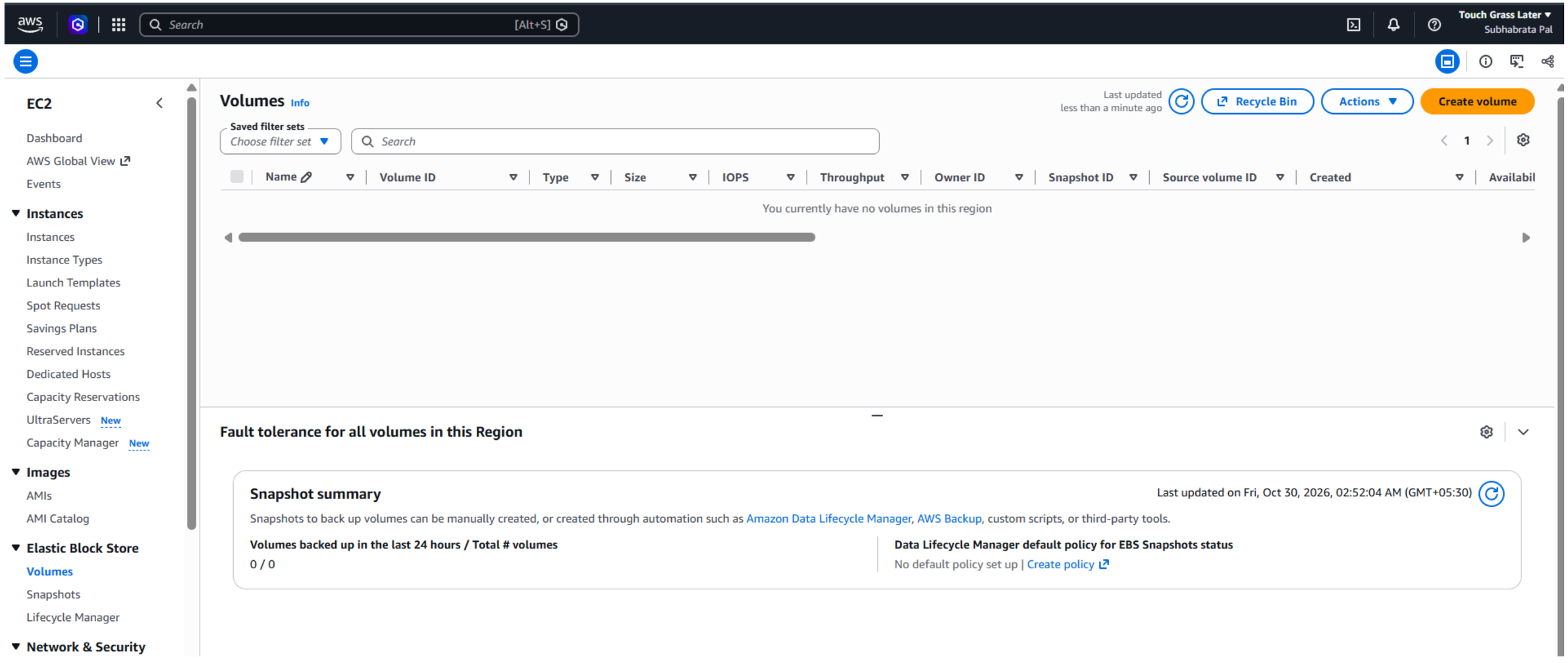
E22 / Cleanup - EC2 terminated

All three project instances show Terminated.



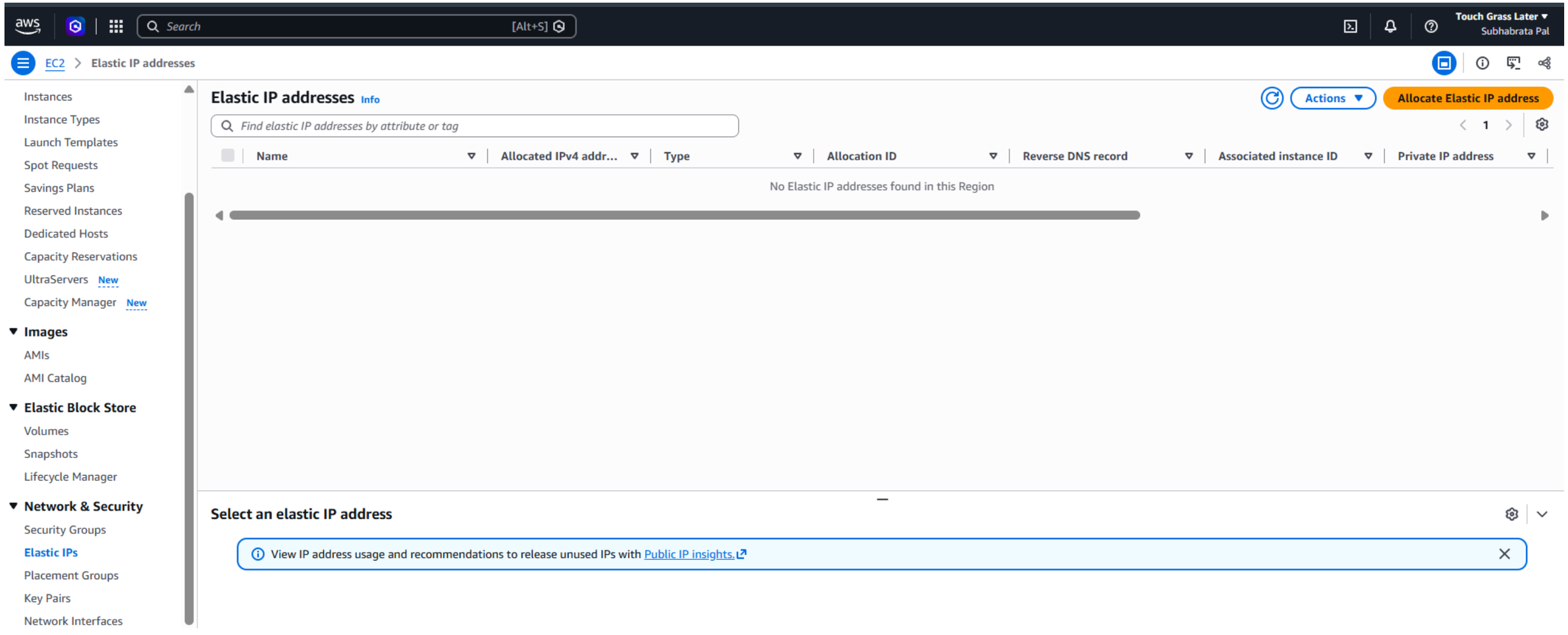
E23 / Cleanup - EBS volumes empty

The regional volumes view states there are no volumes; this is not a dedicated snapshot inventory.



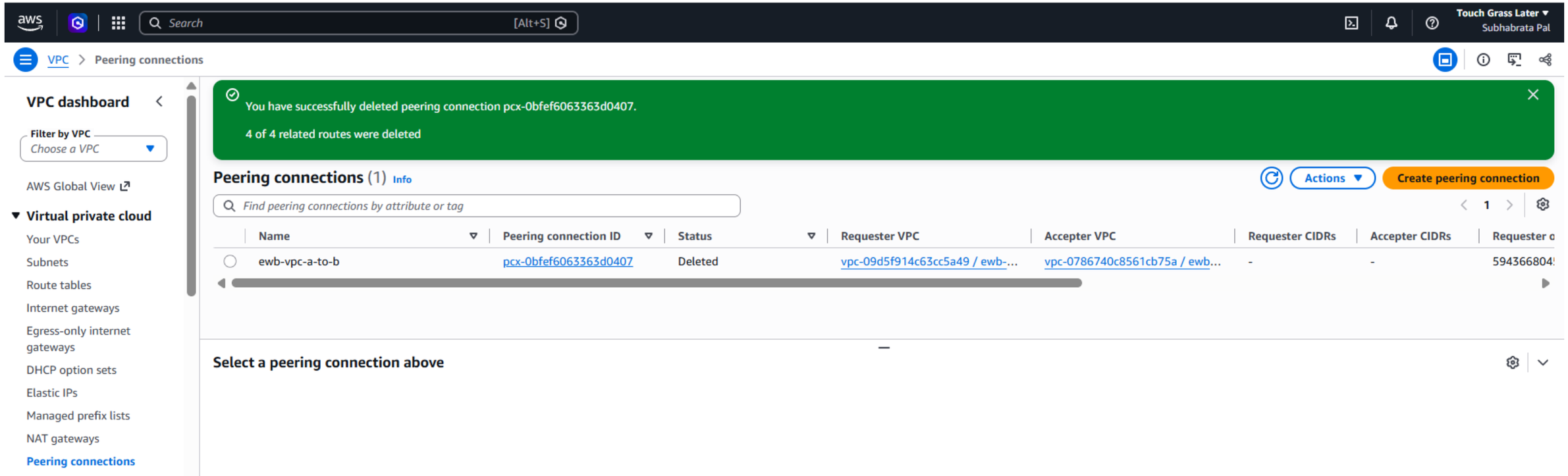
E24 / Cleanup - Elastic IP allocations empty

No Elastic IP addresses are displayed in the Region.



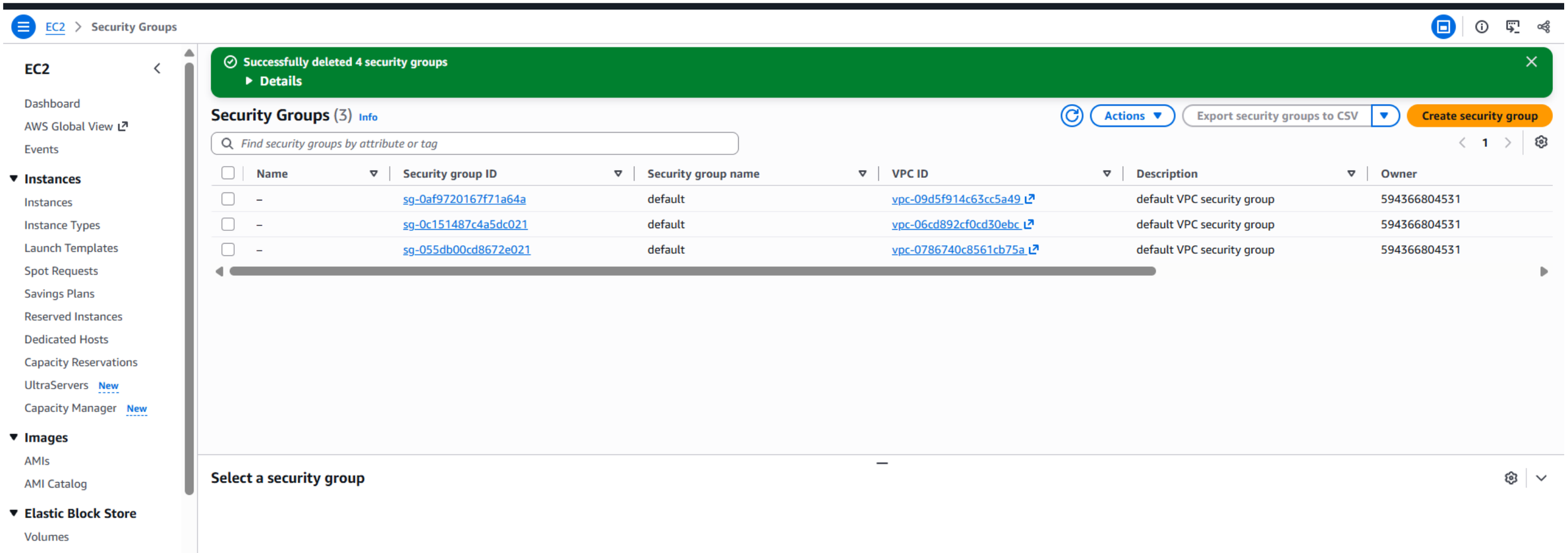
E25 / Cleanup - peering deleted

The peer shows Deleted; the banner states 4 of 4 related routes were deleted.



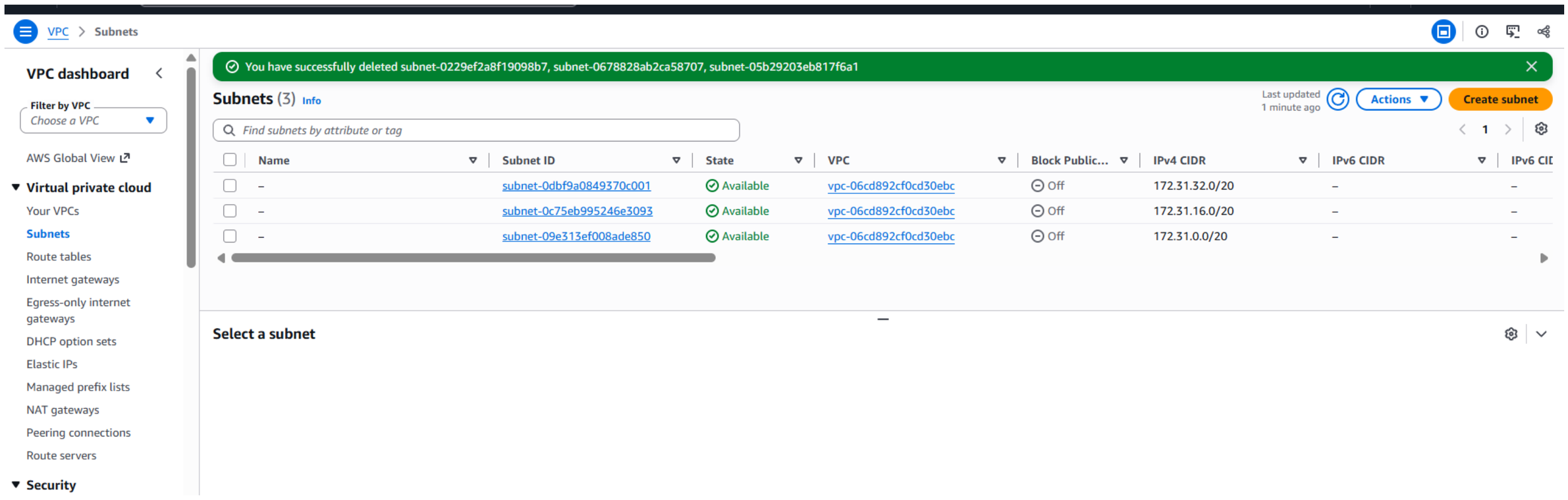
E26 / Cleanup - project security groups deleted

Four security groups were deleted; the remaining rows are default security groups.



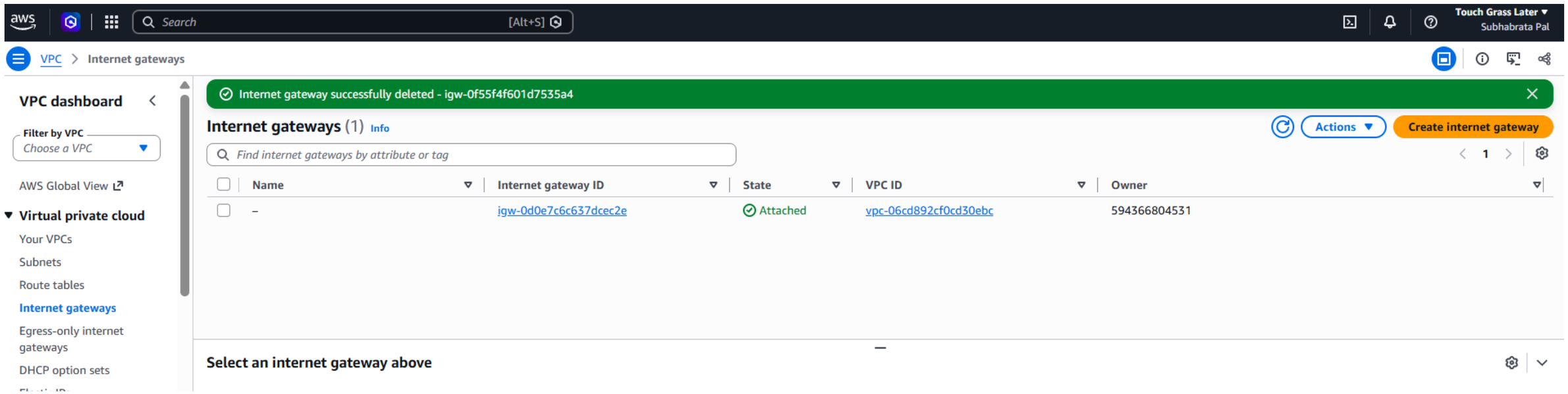
E27 / Cleanup - project subnets deleted

The banner names all three project subnet IDs. Only default-VPC subnets remain.



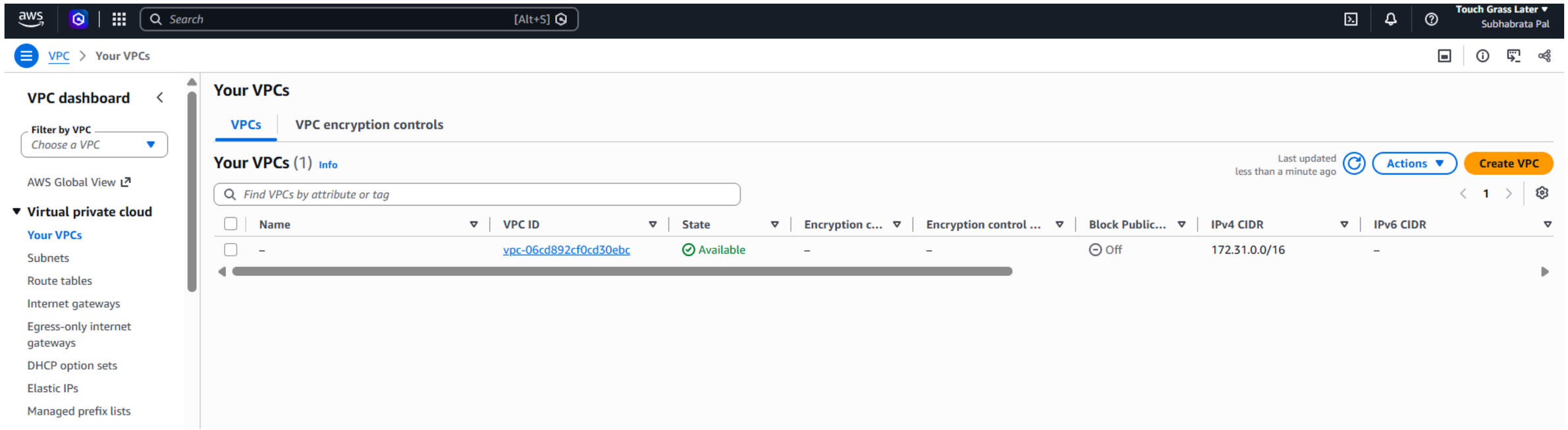
E28 / Cleanup - project internet gateways removed

The displayed list retains only the default VPC gateway; a project gateway deletion is confirmed.



E29 / Cleanup - project VPCs removed

Only the default 172.31.0.0/16 VPC remains in the displayed VPC list.



E30 / Cleanup - EC2 key pair deleted

One key pair was deleted and the list is empty. Local private-key removal is not shown.

